Sanoj PD. Dahal

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Education

IOE, Purwanchal Campus, Dharan **Sunsari, Nepal**
Bachelor of Computer Engineering *Aug 2020 – May 2024*

- **Relevant Coursework:** Data Structures, Design and Analysis of Algorithms, Big Data, Object-Oriented Programming, Database Management Systems, Professional Ethics

Technical Skills

Programming Languages: Python, SQL
Data Science & ML: NumPy, Pandas, Scikit-learn, Imbalanced-learn, XGBoost, Feature Engineering, Model Evaluation
Deep Learning & NLP: TensorFlow, Keras, PyTorch, LangChain, LangGraph, Transformers, Computer Vision (PoseNet)
Data Visualization: Matplotlib, Seaborn, Power BI
Tools & Platforms: Git, Jupyter Notebook, VS Code, Google Colab, MongoDB

Projects

Customer Churn Prediction & Analysis | *Scikit-learn, Pandas, XGBoost, SMOTE*

- Developed a machine learning model to predict customer churn for a telecom company, performing comprehensive EDA, feature engineering, and handling class imbalance using SMOTE.
- Trained and compared Decision Tree, Random Forest, and XGBoost classifiers, using cross-validation and hyperparameter tuning (RandomizedSearchCV) to optimize model performance.

Explainable AI for Nepali Character Recognition | *Machine Learning, Deep Learning, Computer Vision*

- Built and trained a CNN-based model for Nepali character recognition.
- Applied Grad-CAM (Explainable AI) to visualize model decision regions.

Content-Based Movie Recommendation System | *Machine Learning, NLP, Recommendation Systems*

- Built a content-based movie recommendation system using TF-IDF and cosine similarity on metadata from 5,000+ movies.
- Achieved user satisfaction by delivering accurate and personalized recommendations during evaluation.

Autism Spectrum Disorder Prediction System | *Machine Learning, Classification, SMOTE*

- Developed a classification system using behavioral features and demographic data with comprehensive EDA and feature engineering.
- Achieved 93% cross-validation accuracy with tuned Random Forest model using SMOTE for class imbalance and RandomizedSearchCV for hyperparameter optimization.
- Implemented and compared Decision Tree, Random Forest, and XGBoost classifiers with 5-fold cross-validation.

RAG-Based YouTube Question Answering Chatbot | *LLMs, LangChain, Embeddings, Vector Databases*

- Built a Retrieval-Augmented Generation (RAG) chatbot using LangChain to answer questions over YouTube video transcripts.
- Improved response accuracy and reduced hallucinated responses compared to baseline LLM outputs.